Md Ashfaqur Rahman

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EDUCATION

Khulna University of Engineering & Technology (KUET), Bangladesh
Bachelor of Science in Computer Science and Engineering — Jan 2020 – Sep 2025
CGPA: 3.32 / 4.00

RESEARCH INTERESTS

- **Machine Learning**
- **Deep Learning**
- **Computer Vision**
- **Natural Language Processing**
- **Human-Computer Interaction**

RESEARCH EXPERIENCE

Undergraduate Thesis — Feb 2024 – Aug 2025
Supervisor: **Dr. Muhammad Sheikh Sadi**
Title: *Bridging the Accessibility Gap: A Comprehensive System for Arabic Braille Interpretation*

- Curated and extended an Arabic Braille dataset by collecting samples for underrepresented characters, diacritics, and ligatures, addressing class imbalance in low-resource settings.
- Designed a two-stage recognition pipeline using YOLOv8 for Braille cell detection followed by a custom CNN-based classifier for character recognition.
- Conducted comparative evaluations against existing state-of-the-art approaches, achieving superior accuracy and robustness on complex Braille sequences.
- Developed a language-aware post-processing module to reconstruct syntactically valid Arabic sentences from noisy character-level predictions.
- Performed systematic experimentation including ablation studies, error analysis, and model benchmarking for academic publication.

PUBLICATION

Ashfaqur Rahman and Muhammad Sheikh Sadi, "A Comprehensive System for Arabic Braille Interpretation," *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 2992-2997, doi: 10.1109/ICCIT68739.2025.11491613.

TECHNICAL SKILLS

Programming: Python , C, C++, Java, SQL
Machine Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, skfuzzy, spaCy, Matplotlib
Deployment Tools: Git, Docker
Libraries & Frameworks: Laravel, OpenCV.
Technologies: Generative AI, Data Management, Deep Learning, Computer Vision, NLP
Others: Linux, LaTeX.

SELECTED PROJECTS

- **Medical Chat SOAP Summarization**
  Description: Description: Fine-tuned Flan-T5-base with LoRA (r=16, 1.77M trainable parameters, 0.71%) to summarize medical dialogues into SOAP notes on a 10,000-sample dataset (9,250/500/250 train/val/test; 3 epochs, 45 min on a T4). Test results: ROUGE-1 0.503, ROUGE-L 0.369, BERTScore F1 0.886.

- **Mancala Game - AI vs Player**
  Description: Developed an AI-powered Mancala game using Pygame, integrating Alpha-Beta pruning for optimal AI moves, Genetic Algorithm for human move suggestions, and Fuzzy Logic to compute real-time winning probabilities based on board state.

- **News Brief-A News Summarizer App**
  Description: Developed an Android application integrating Python-based NLP pipelines to perform extractive news summarization using the TextRank algorithm.

- **3D Football Stadium**
  Description: Developed a real-time 3D football stadium simulation using Modern OpenGL 3.3, implementing programmable-pipeline rendering with custom GLSL shaders, Phong illumination, and texture-mapped geometry. Integrated physics-based ball trajectory computation, multi-light models, and fully interactive camera/navigation controls.

- **School Management System**
  Description: Built a full-stack Laravel-based School Management System implementing RBAC security, normalized relational schema design, RESTful CRUD services, and automated academic processes across structured relational entities.

- **Enterprise Banking Database Design and Implementation**
  Description: Designed and implemented a fully normalized banking database with complex constraints, multi-entity relationships, and advanced SQL/PLSQL operations.

- **Compiler Project**
  Description: Developed a compiler front-end for a custom high-level language using Flex and Bison, supporting variables, expressions, control structures, functions, and I/O. Implemented semantic analysis, symbol-table management, and runtime interpretation to execute programs end-to-end.

- **CardiacRecorder – Android app for real-time cardiovascular monitoring.**
  Description: Developed an Android application for real-time acquisition, validation, and visualization of cardiovascular metrics using a modular MVVM architecture. Implemented automated unit/UI testing and GitHub Actions CI workflows to ensure reliable performance, scalable feature integration, and maintainable code quality within a collaborative development environment.

ACTIVITY

- **Event Organizer, BitFest 2025**

  Organized the Inter-University Programming Contest (IUPC) at the 3rd KUET CSE National Festival.

  Skills: Team leadership, event coordination, cross-team communication

LANGUAGES

Bangla: Native
English: Proficient